

EMAN ASHRAF | SOFTWARE ENGINEER

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About Me

Bachelor of Science in Computer Science from Ain Shams University with hands-on experience in Python, Java, C++, PHP and C. Passionate about leveraging technology to solve real-world problems. Proven track record in developing efficient system architectures and machine learning applications.

Education

Ain Shams University | Cairo, Egypt

October 2022 – August 2026

Bachelor of Science in Computer Science

Projects

Beyond Earth – Cosmic Journey | *FastAPI, Python, JS, HTML, CSS*

september 2025 – October 2025

- Developed the backend of the Beyond Earth – Cosmic Journey project using FastAPI, enabling seamless data flow between the interactive 3D frontend and scientific exoplanet datasets.
- Implemented secure OTP-based email verification to enhance authentication and safeguard user access.

University Management System | *PHP, Laravel, MySQL, Blade*

september 2025 – september 2025

- Developed a Laravel-based CRUD web application with MySQL integration, implementing full create, read, update, and delete functionalities.
- Implemented user authentication and role-based authorization, ensuring secure access control for different user levels.
- Applied form validation and structured backend logic to enhance data integrity and provide seamless front-end to back-end communication.

Faster | *HTML, CSS, JS, Bootstrap*

september 2025 – september 2025

- Developed a fully responsive website using HTML, CSS, JavaScript, and Bootstrap
- Strengthened front-end development skills by focusing on clean, efficient code and structured design practices

Map Routing System | *C#, Windows Form*

April 2025 – March 2025

- Engineered a weighted graph from real-world map data with over 200,000 nodes, supporting both pedestrian and vehicle routing.
- Optimized Dijkstra's algorithm with priority queues, reducing query time by 60% for large-scale routing scenarios.
- Implemented a responsive Windows Forms interface to dynamically visualize shortest paths and travel modes.

VideoGame Sales On Steam Prediction | *Python, Machine Learning*

April 2025 – March 2025

- Built regression and classification models to predict game sales, achieving up to 90% accuracy through extensive hyperparameter tuning.
- Cleaned and engineered features from 91,000 game records including bundles, and genres.
- Engineered features and performed data cleaning across diverse sources (base games, DLCs, demos, sales) to enhance dataset quality and enable effective analysis of user engagement and sales patterns.

FOS - A Simple Operating System | *C, Assembly*

October 2024 – December 2024

- Produced dynamic memory allocation (First Fit strategy), reducing fragmentation by 30% compared to baseline.
- Planned dynamic memory management for both kernel and user heap, utilizing efficient allocation strategies such as FirstFit and paging-based allocation to optimize performance and reduce fragmentation.

Skills Summary

Programming Languages : Python, Java, C, C#, PHP, JavaScript, C++, TypeScript, Scala

Programming Concepts : Object-Oriented Programming (OOP), Data Structures, Algorithms

Tools: Git/GitHub, MySQL, HTML/CSS

Technologies/Frameworks: FastAPI, Laravel, Bootstrap

Languages : Arabic (Native), English (Professional Proficiency)

EXTRACURRICULAR ACTIVITIES

APPLAi Student Chapter

October 2024 – Present

- Committee Member** Contributed to computer vision projects, implementing deep learning models for image processing, object detection, and recognition tasks within APPLAi initiatives.